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Mechatronic Actuators Trainer

Fundamental Labs – Brushless DC Motor

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

FCC Notice This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Industry Canada Notice This Class A digital apparatus complies with CAN ICES-3 (A). Cet appareil numérique de la classe A est conforme à la norme NMB-3 (A) du Canada.
Japan VCCI Notice This is a Class A product based on the standard of the Voluntary Control Council for Interference (VCCI). If this equipment is used in a domestic environment, radio interference may occur, in which case the user may be required to take corrective actions.

この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。 VCCI–A

Waste Electrical and Electronic Equipment (WEEE)
This symbol indicates that waste products must be disposed of separately from municipal household waste, according to Directive 2012/19/EU of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces the environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources.

电子信息产品污染控制管理办法 (中国 RoHS)

中国客户Quanser Consulting Inc. 关于关于限制在电子电气设备中使用某些有害成分的指令 (RoHS)。

CE Compliance

This product meets the essential requirements of applicable European Directives as follows:

- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.

Brushless DC Motor Lab

Why Explore a Brushless DC Motor?

There are two types of commonly used DC motors, this lab will help you explore Brushless DC (BLDC). BLDCs are efficient and more durable than regular DC motors. As described in their name, they don't use brushes or a commutator to deliver current to the coils, instead, they use a rotating permanent magnet and stationary coils. This design reduces wear, which makes them last longer, as well as reducing electrical noise while still allowing precise control of torque and speed. By not having brushes and using electronic switching of current, BLDC motors are able to maintain high torque throughout rotation. As such, these motors are highly power efficient, leading to lower energy losses and less heat production.

BLDC's efficiency and reliability make them ideal for applications that require precision and continuous operation, for example, appliances such as washing machines, air conditioners, vacuums, and hard disc drives. However, these motors are harder to control as the actuation is achieved through the interaction of the permanent magnet rotating based on the direction of the generated magnetic field in the stationary coils.

Background

This lab is part of the Fundamentals labs for the Mechatronic Actuators Trainer. These labs will guide you through four different types of motors, so you can understand their operating principle and get you comfortable with using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**004_motor_bldc**.
- 4_concept_reviews/Mechatronics/**007_rotary_sensors (rotary encoders section)**.

Getting Started

In this lab, you will use the **Actuators Trainer** to learn about the basics of **brushless DC motors**. The lab will guide you through the usage of hall sensors in the motor and the associated logic needed to drive the motor. You will experiment with this to understand how to drive the motor at different speeds and directions. You will also be comparing your results to values found in the datasheet.

Before you begin this lab, ensure that the following criteria are met:

- Make sure you have either the provided Brushless DC motor, or another BLDC motor properly wired to be able to interface with the Actuators Trainer.
- The provided BLDC motor is this one. Confirm your motor is wired properly to the 3-pin connector to run it and the 5-pin encoder cable.

- Make sure the Actuators Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/actuators_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.